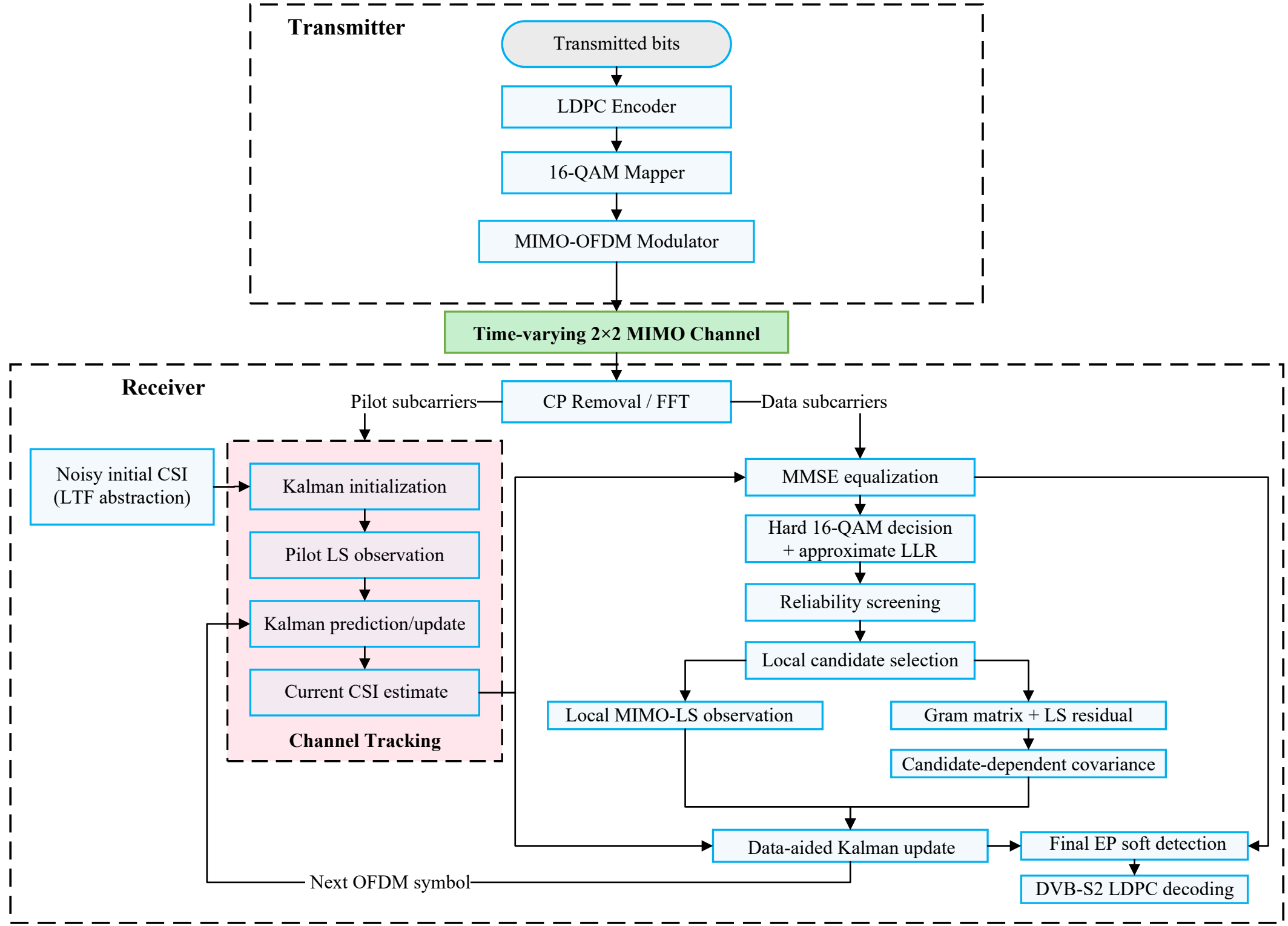

Transmitter
Transmitted bits
LDPC Encoder
16-QAM Mapper
MIMO-OFDM Modulator
Time-varying 2×2 MIMO Channel
Receiver
CP Removal / FFT
Pilot subcarriers
Data subcarriers
Noisy initial CSI
(LTF abstraction)
Kalman initialization
Pilot LS observation
Kalman prediction/update
Current CSI estimate
Channel Tracking
MMSE equalization
Hard 16-QAM decision
+ approximate LLR
Reliability screening
Local candidate selection
Local MIMO-LS observation
Gram matrix + LS residual
Candidate-dependent covariance
Data-aided Kalman update
Final EP soft detection
DVB-S2 LDPC decoding
Next OFDM symbol